

OUTPUT EXPLANATION

Answer to question 1: Distribution of Continuous Variables (Distribution_table_1.csv)

I used the Shapiro-Wilk test to check the distribution of continuous variables for each outcome category on its own. A p-value greater than 0.05 signified a normally distributed variable, while a p-value less than 0.05 denoted a non-normally distributed variable.

The results showed that only a few variables across outcome groups looked like a normal distribution. Most of the variables were not normal, which means they were skewed or had outliers. This mixed pattern made it okay to use different summary statistics based on how the variable was spread out.

Answer to question 2: Descriptive Statistics by Outcome (Descriptive_table_1.csv)

The descriptive statistics table revealed substantial differences in central tendency and dispersion across result groups for several variables. Some variables showed clear differences, which could be related to the result, while other variables showed similar patterns across groups.

EXTRA POINTS

Question 1

A study of the dataset found that the "lipids5" variable has some values that are missing. To fill in the gaps, the median value was used. This method is especially good for skewed biomedical data because it can handle outliers and keeps all observations without getting rid of any data points.

Question 2 and 3: (Extra_Distribution_table_2.csv and Extra_Descriptive_table_2.csv)

After imputing the missing values in `lipids5`, the distribution analysis and descriptive statistics were re-run. The updated results demonstrated that including the imputed `lipids5` values did not substantially alter the distribution patterns or overall trends observed in the other variables

Conclusion

The distribution analysis and descriptive statistics were run again after filling in the missing values in "lipids5." The revised results indicated that the distribution patterns and general trends identified in the other variables remained largely unaffected by the incorporation of the imputed `lipids5` values.